

research laboratories to improve the performance of these batteries.

The global metal-air battery market size was valued at \$445 million in 2021 and is projected to reach \$745 million in 2028, with iron-air batteries potentially holding about a 20% share of the total global market by 2030. Fig. 15 shows the projected market share of various types of metal-air batteries in 2028.

Cost analysis

Form Energy is a company that has built a \$760 million factory on 55 acres of land in West Virginia, USA. They have improved this 140-year-old technology and are likely to start commercial production of

iron-air batteries in their new facility in 2024. Current estimates indicate that iron-air batteries could cost around \$25 per kWh capacity, compared to up to \$200 per kWh for Li-ion batteries.

Basis of cost estimate for iron-air battery:

Plant capacity: Manufacturing 100,000 packs/year

Designed power: 110kWh/pack

Open circuit voltage: 1.36V

All items are outsourced, and the plant assembles the same to produce battery packs.

Cost break-up:

1. Cost of material and hardware: \$12/kWh

This includes active material and current collectors for negative and positive electrodes, carbon, binders, electrolyte, separator, battery jacket, all hardware, etc.

2. Cost of air delivery system comprising air compressor, pressure swing adsorption system to concentrate O₂, piping, instrumentation, electrics, etc: \$7/kWh

3. Cost of pack integration: \$3/kWh

4. Labour cost, sales promotion, administration, overhead, R&D, depreciation, warranty, profit, etc: \$3/kWh

5. Total unit cost (addition of 1 to 4): \$25/kWh

The cost breakdown in circular template is shown in Fig. 16.

Iron-air batteries are not a replacement for Li-ion storage systems but rather a complement to them. Li-ion batteries have a high power-to-weight ratio or energy density while maintaining a small size, which is ideal for EVs. They are also great for fast response actions like frequency regulation and grid stabilisation.

In contrast, iron-air batteries would be best at storing and supplying large amounts of energy at lower power and density. They are designed to be hooked together in grids. The same chemical process that ruins a steel structure, if not painted, could help power the electric grid in the future and may save the planet from doomsday due to climate change. It is the process of reversible rusting: while rusting the iron electrode, the battery discharges, and while charging the iron-air battery, we are pushing electricity back in. The battery is unrusted.

There is enough cheap iron available on our planet and utilising it in the battery makes it possible to keep electric grids running 24 hours a day on intermittent, but inexhaustible solar and wind energy. The 140-year-old technology in a new avatar can provide backup energy to power electric grids not just for days, but months. However, getting there would require more research and development work. It is expected that within a few years, commercial production of this type of battery will start. **EFY**

Concluded

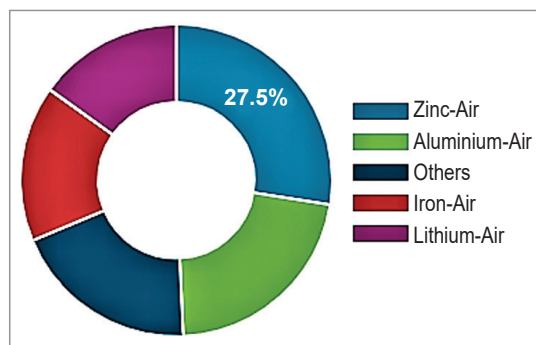


Fig. 15: Projected market share of metal-air batteries in 2028 (Source: www.fortunebusinessinsights.com)

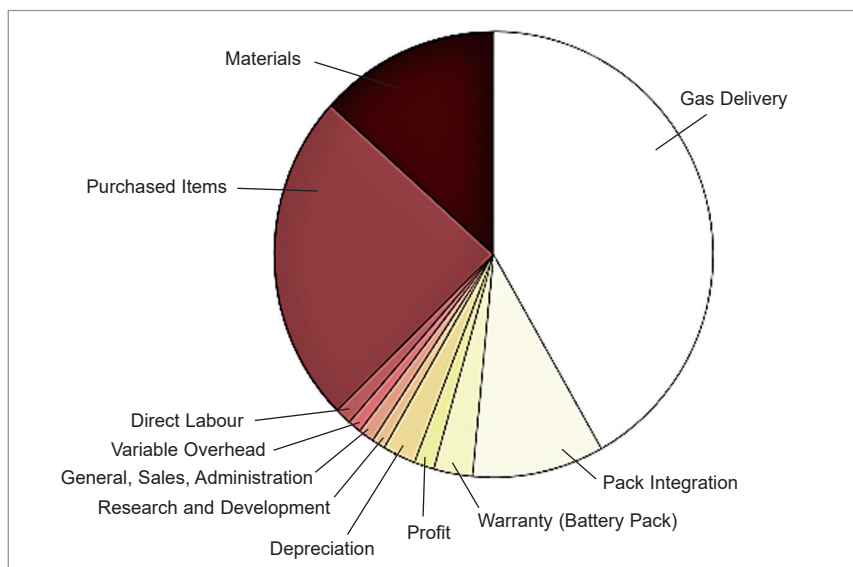


Fig. 16: Break up of total unit cost of iron-air battery